Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 19.5 AV Summary

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

UE26ME142A Unit 2 Class 19.5 AV Summary

1 UE26ME142A Unit 2 Class 19.5 AV Summary

Study focus: UE26ME142A_Unit 2_Class 19.5_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000170 · EV0000171

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Svinkitish from the Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number twenty eight.
- We are in the process of solving few problems.
- Under chapter three, there is equilibrium chapter.
- And probably this would be the last class in this chapter.
- In the coming classes I will be starting with the new chapter, that is chapter number four, that is process.
- under equilibrium chapter.
- The authors are Miriam, Craig and Bulton.
- The statement of the problem goes something like this.
- The 50 kg point exerts a pull of 150 newton on the row.
- Rig the ashton.
- Compute the force supported by the pin at the hinge O.
- is hundred kg and we all know that the sulphate is always going to be acting at the center of the total span or total distance of the beam.
- The distance at which this 100 kg into 9.81 newtons will be acting is at a distance of 4 by 2, which is equal to 2 meters.
- And also he says that the weight of the boy who is standing on the beam is 50 kgs.
- It will go to act at the center of gravity of the boil, which is indicated in the diagram as 1.5 meters from the left support, that is this distance where the weight of this person or the boy

is gonna act that is fifteen to nine point eight one kgs will be acting at this point.
- And it also says that the boy exerts a pull of 150 newtons.
- Whereas at this point it will be acting upwards.
- Since the boy is standing on the beam, the weight of the boy is going to be acting on the beam.
- Whereas when he tries to pull this rope, the tensile force always going to be in pair, acting towards each other.
- However there will be force in this cable also.
- Again here it will be opposed to this which will be acting.
- Like this, from moving away from here, here it will be moving away.
- We will have a force here which will be acting downwards.
- We will have a force here which will be acting downwards.
- In addition to this, we will also have a roller here at point A.
- Then we have a pin or an inch connection at O.
- What are the forces that's going to be acting on this beam now as two reaction at O, one vertical, one horizontal, one reaction at A which will be acting up vertically upwards.
- We have the.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 05:00–09:59

Tension force in this cable which is acting downwards of magnitude 150 newtons.

Key details

- Here also there will be one more cable which will be acting downwards of magnitude 150 newtons, and we will have the weight of the which is fifteen to nine point eight one.
- In addition we will have also hundred into nine point eight one will be active in the center of the beam.
- The forces in this cable will balance with each other, remember this carefully.
- Or what is that we have to find out?
- We have to compute the force supported by the pin at the end.
- that is O.
- You take the momentum out EA.
- Or movement about O because that's where you have the unknown support reaction that is acting.
- If you take the moment about A, you get the reaction acting at O, vertical reaction into 0.75, I assume it to be acting upwards, it will be anti-clockwise.
- Then the sulfate of the beam which will be acting.
- But the center of magnitude hundred into nine point eight one.
- But from A it is going to be at a distance of 1.05.
- Then the force in this will also act in the anticlockwise moment, which will be equal to 115 to 2.5.

- Then we will have the force that is acting in this cable which is also equal to 150 newton into force is 3.25.
- In addition we will also have the weight of the boil which is 15 to 9.81.
- From this end if I want to find out it will be equal to 3.25 minus 1.5, which is going to be 1.75.
- And that vertical reaction we will find out.
- free body diagram I'm not sure it is not visible here, however I have told you.
- That is at the point where the roller is acting, as I have told you, and one fifty is the force in the cable into 3.25 that is the extreme end left cable, then one more cable which is acting 1.15 into 2.5 plus the sulphate of the beam.
- Should be acting at the center from A it is 1.25 plus the weight of the boy which is 15 to 9.8 one acting at it.
- All this are clockwise, so it is taken as positive minus the reaction acting at the hinge into the distance which is 3.25.
- However, if you take clockwise as positive and anti-clockwise as negative, or clockwise as negative and anti-clockwise as positive, you get the same equation.
- And that's what I've told you.
- Force in the cable is not considered because that will be acting opposed to each other, that will balance with each other.
- And also there is no other horizontal reaction that is acting, that's why the reaction at.
- O along the horizontal direction is a zero.
- So we move on to the next problem.
- Next problem is 3 bar 54.
- The pin A which connects the 200 kg steel beam with center of mass at the G to the vertical column is welded both to the beam and to the column.
- To test the weld.
- The eighty kg man loads the beam by exerting a eighty newton force on the row, which passes through a hole in the beam as shown, calculate the chart.
- or couple yum supported by the pin.

Example or application from the lecture

- Let's move on.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 10:00–14:59

There is a pin at here, that is this one.

Key details

- Then he says that the beam is welded.
- To the column.
- To test this weld, because without the weld, the beam would try to rotate in the clockwise direction.

- Since there is a weld there.
- This would resist the rotation.
- And he also says that the person exerts a 800 newton force on the rope, which is passing through the beam by a hole, which is drilled.
- The same rope is continued here.
- Here also you will have the 800 newton force.
- And that is taken by this pulley.
- But however if I cut it here, there will be 1804 newton which will be acting downwards, plus here also.
- Because of this is asking calculate the target M or the couple M.
- Supported by the pin that is at this point what is the moment that is being supported or the target that is being supported.
- You have the reaction R here.
- The horizontal reaction is not indicated because there is no horizontal force here.
- Then you have a momentum that is indicated in the anticous direction.
- Cell weight of the beam, cell feed of the person, then the cable weight.
- Which will be acting on the person itself, even though there is a hole there, it is not going to be directly acting on the beam.
- It is going to be transferred from the.
- Plus, in addition to that, you will have also the cable which is attached to this point, so you will also have that force.
- The force acting is.
- The sulfate of the bee?
- Self rate of the person.
- As I've explained.
- We get the s first sulfate of the person eighteen to nine point eight one.
- The distance at which it is acting is thousand eight hundred from the left end, that is A.
- Plus we have one hundred.
- Kg that is 29.81 into thousand two hundred is the sulfate of the beam that is taken here plus the cable whatever is there that has been taken here.
- Only one unknown is M and this is acting in the.
- anti-clockwise direction, all the clockwise forces or the moments are taken positive, anti-clockwise moment is taken as negative.
- I hope all of you have got this.
- I have a doubt of whether the force that is acting.
- Or the force with which the person is pulling is not 800, I think it is 300.
- Make that correction, please.
- It is 300 newtons.
- No, sorry, it is kilo Newton meetings.

Example or application from the lecture

- The sulphate of the beam is given as two hundred kgs, which will be acting at the center, that is given as G here, which is at a distance of thousand two hundred mm from the left hand.

- You can see here, they have given that it is welded here.
- And they have also given you the weight of the person which is ADKG, which will be acting along the center line of the person, which is at a distance from A by 1200 plus 600, which is 1800.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 15:00–19:59

We have come to the end of equilibrium chapter.

Key details

- So basically what we have to understand in equilibrium chapter is first thing is the equations of equilibrium.
- For a two dimensional there are three equations of equilibrium.
- One is sigma.
- F along the vertical, algebraic sum of all the forces acting along the vertical is zero.
- Algebraic sum of all the forces acting along the horizontal is zero, then algebraic sum of the moment at any point should be equal to zero.
- They are called as determinate structure.
- If you have more than three constraints,.
- Then the number of constraints more than three is called as a redundant reaction or constraint.
- And we need to have more equations to solve such problems other than the three equations of economy.
- Such structures are called as indeterminate structure.
- Then we should also understand what are the different types of supports that we have.
- Smooth surface, rough surface, roller surface, or roller support, then sliding guide, then pin connection, building connection, gravitational force, linear, spring,.
- And torsional spring.
- Then how to draw the free body diagram.
- Then what are the different types of supports that we have?
- How do we replace these supports by reaction.
- With this we will be able to solve the problems in case of equilibrium chapter.
- In the next class, I'll be starting with chapter four, which is on trusses.
- Again, in trusses, we will be continuing to use these equations of equilibrium along with what we have learnt in unit one, that is resolution of forces.
- Moment topic, couple as well as the resultant.
- Knowledge is very much required to understand chapter four, that is process chapter.
- Okay, along with that you should also understand other things related to trusses which I will be discussing more in depth when I go to take it in the next class.
- Whatever problems I have discussed here, there are other problems also which I would request you to go through and solve them.
- I'll be very happy to help you.

- We have come to the end of today's class as well as end of chapter three, that is equilibrium.
- My name is Dr.
- Sankitesh from Civil Engineering under the Department of Science and Humanities.
- That is class number thirty.
- However, we have one more class that is 29, which is kept for experiential learning.

Example or application from the lecture

- I have explained this, which is given also in the textbook on page number one hundred and eleven if I'm not wrong, and as well as hundred and twelve, where we have discussed about cable.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 19.5_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/019_UE26ME142A_Unit_2_Class_19.5_AV_Summary__145da9b4503db2ea.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.

9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0057. Sidecar variants compare...

C Revision and retrieval practice